RNGD • CONCEPT-TO-SCHEMATIC AI

RNGD AI Studio

Turning a building request into a working design — automatically

A few AI agents read a request; one fast tool turns sliders into a real 3D building.

The problem: building design starts slow

Today, every project starts from scratch.

- A customer's wish list has to turn into a real site plan and floor plan
- Checking zoning and building-code rules is manual, page by page
- Problems — like parking that doesn't fit — often surface late, after real design work is done

RNGD's goal

Use AI to speed up the very first step: **concept to schematic design.**

We built two tools, in one app

1. Multi-Agent Pipeline

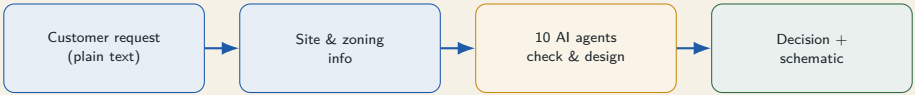
A written customer request goes in. Ten AI agents check it against the site, the rules, and each other — and hand back a full schematic *plus a decision*.

2. IFC Design Studio today's focus

Move a few sliders — units, floors, parking type — and get an instant, real 3D building model, checked against the rules as you go.

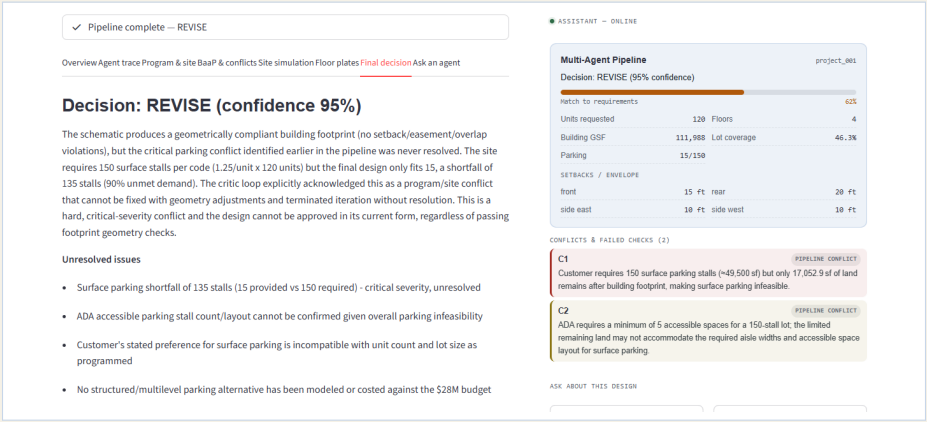
Both share one knowledge base and one chat assistant that can answer questions about either.

Pipeline: how it works, in four steps



- Agents include: parsing the request, computing the buildable envelope, matching RNGD's building-module catalog, a physics-style site simulation, and a computer-vision quality check
- One final AI call reads everything and makes the call: approve, revise, or "not enough information"

Pipeline: it doesn't just say yes



For a 120-unit request on this sample site, the AI found a real problem: the lot cannot fit the required 150 parking stalls, so it **REVISE**d the project instead of approving it — with the exact numbers and reasons.

Now: design in real time

The IFC Design Studio

Move a slider. Get a real building.

How the Design Studio works, in three steps

1. Tell it what you need

Units, floors, unit mix, parking type, zoning preset — a few sliders, or paste a request and let AI fill them in.

2. It checks the rules

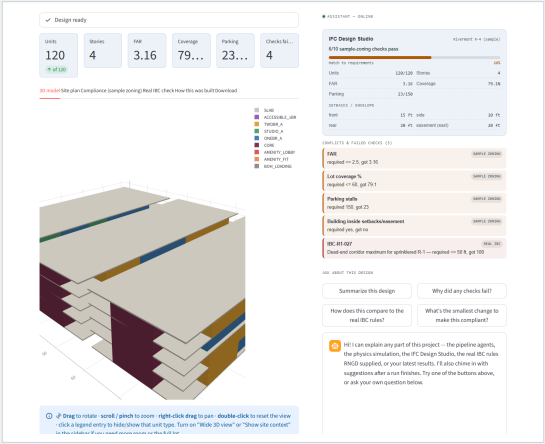
A real buildable envelope, a sample zoning code, and RNGD's **real** 2021 building-code data, all at once.

3. You get a real building

A standard, open **IFC** 3D file — the same format real architecture software uses — instantly.

No AI model is required for this part — it's fast, repeatable, rule-based engineering, so it updates the moment you move a slider.

See it in 3D



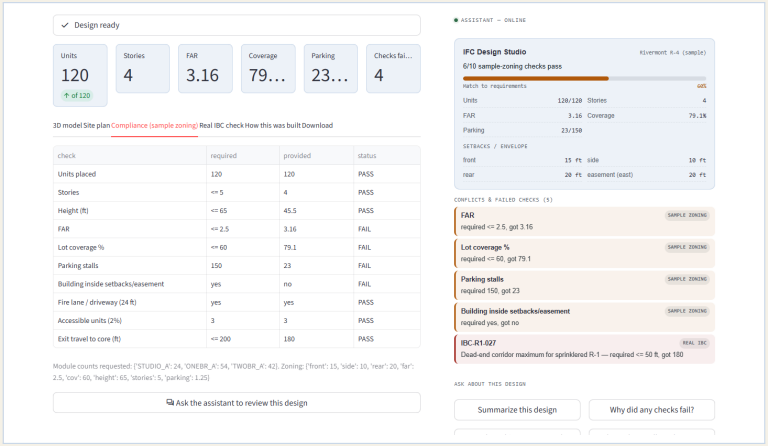
Every unit type, stair core, and amenity is color-coded and placed automatically from RNGD's building-module catalog — drag to rotate, scroll to zoom, right in the browser.

Where everything sits on the lot



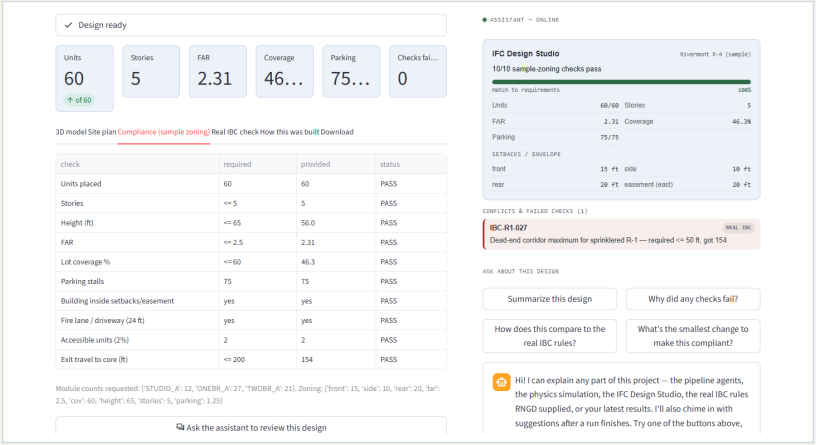
Blue = building, yellow = utility easement, dashed red = the buildable limit, grey = parking. This particular design only fits 23 of the 150 parking spots this program would need.

The AI checks its own work



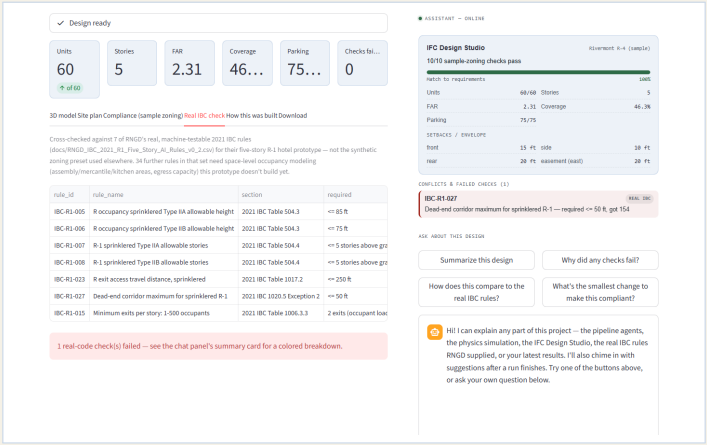
Every design is scored against ten real zoning rules. Red **FAIL** rows mean exactly that — the AI never hides a problem to make a design look good.

Right-size the program, and it all passes



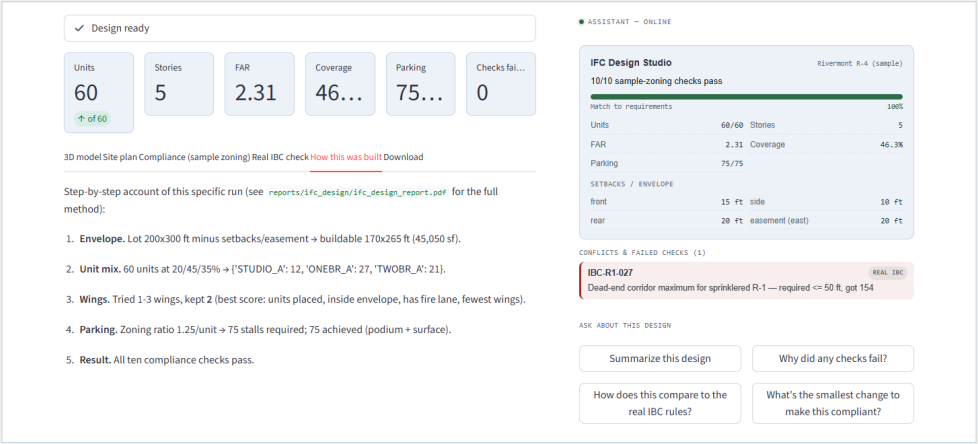
Same tool, smaller program (60 units with podium parking instead of 120 on surface lots): **100% match to requirements**, all ten checks pass. The Studio doesn't just flag problems — it helps you find a design that works.

Grounded in RNGD's own real building code



Not placeholder data — RNGD's actual structured 2021 IBC rule set for their five-story hotel prototype. Here it caught something real: this corridor layout fails the real 50-ft dead-end-corridor limit, even though it passed the simpler sample check.

It explains itself, step by step

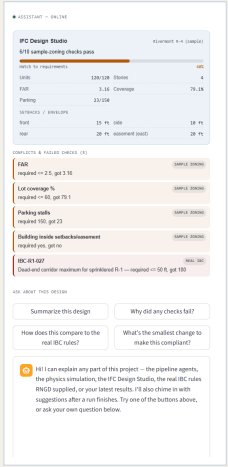


Every design comes with a plain-language account of exactly how the numbers were computed — not a black box.

Ask it anything

A chat assistant sits beside every design and always shows:

- A live, color-coded summary of *this* design
- A "match to requirements" score
- Every setback and failed check, in plain language
- One-click question buttons — no typing required



Why this matters

Speed

Minutes, not days, for a first schematic

Honesty

Flags real problems instead of hiding them

Grounded

Checked against RNGD's own real code data

Try it:

```
streamlit run src/app.py
```

All zoning presets and the BaaP catalog are sample data; the IBC rules and ADA standards are RNGD's own. Prototype output — requires licensed professional review.